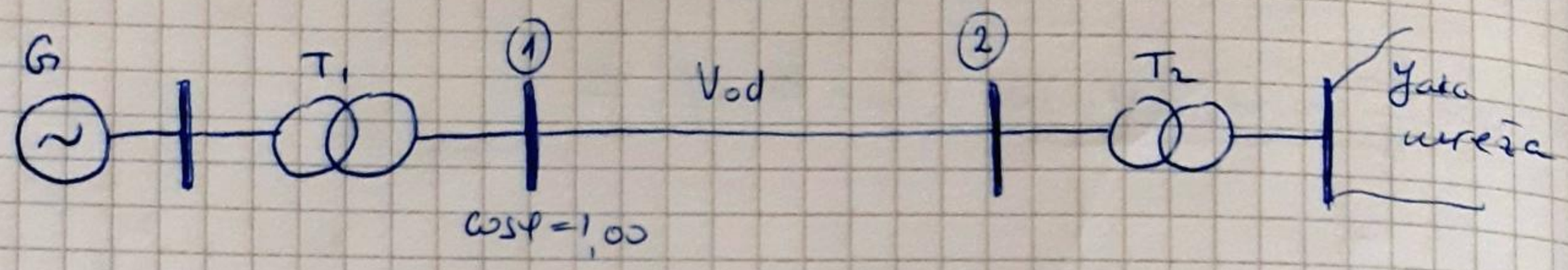
Zadatak za vježbu broj 6



$S_{nG} = S_{nT1} = 500\,MVA$ $\quad U_1 = U_{nV} = 380\,kV$ $\quad S_{nT2} = 500\,MVA$ $\quad U_\infty = const.$

$U_{nG} = 15{,}75\,kV$ $\quad Z_c = 320\,\Omega$ $\quad u_{T2} = 380/220\,kV/kV$ $\quad f = const.$

$u_{T1} = 15{,}75/400\,kV/kV$ $\quad L_v = ?$ $\quad x_{T2} = 10\%$

$x_{dG} = 160\%$

$x_{T1} = 15\%$

Proračun parametara: (380 kV)

$$x_{dG} = \frac{160\%}{100} \cdot \frac{400^2}{500} = 512\,\Omega$$

$$x_{T1} = 0{,}15 \cdot \frac{400^2}{500} = 48\,\Omega$$

$$x_{T2} = 0{,}1 \cdot \frac{380^2}{500} = 28{,}88\,\Omega$$

$Z_c = \sqrt{\frac{l}{c}}$ // karakteristična impedansa,

l, c - podužna induktivnost i kapacitivnost

$\beta = \omega\sqrt{lc}$ // fazna konstanta -> mjera promjene faznog ugla dva fazora U ili I duž voda

$\beta = 0{,}06\,\frac{°}{km}$

a) $L_V = ?$

Da bi sistem ostao stabilan na male poremećaje:

$$P_{syn} = \frac{\partial P_{Gi}}{\partial \delta_i} > 0$$

$$P_{Gi}' = G_{ii} E_i^2 + \sum_{\substack{k=1 \\ k \neq i}}^{n_G} E_i E_k \left[\cancel{G_{ik}} \cos(\delta_i - \delta_k) + B_{ik} \sin(\delta_i - \delta_k) \right]$$

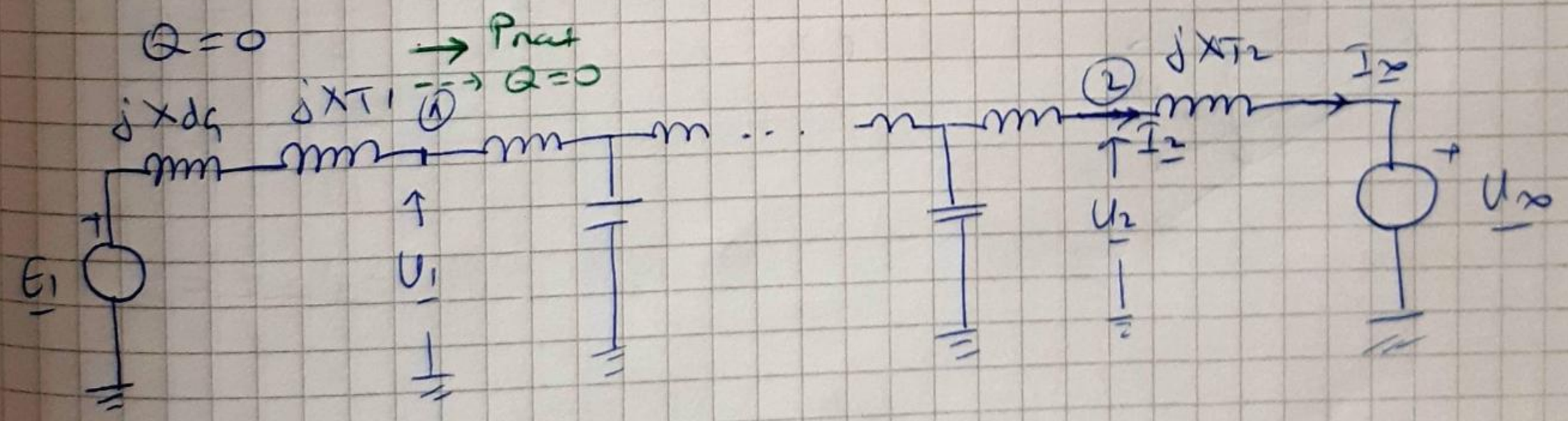
$$P_{G1}' = G_{11} E_1^2 + E_1 U_\infty B_{1\infty} \sin(\delta_1 - \delta_\infty)$$

$$\frac{\partial P_{G1}'}{\partial \delta_1} = E_1 U_\infty B_{1\infty} \cos(\delta_1 - \delta_\infty) > 0$$

$$\delta_1 - \delta_\infty < 90^\circ$$

δ_1 i δ_∞ su fazni stavovi $\underline{E}_1$ i $\underline{U}_\infty$ u odnosu na referentnu osu.

$$P_{nat} = \frac{U_{nV}^2}{Z_c} = \frac{(380\ kV)^2}{320\ \Omega} = 451{,}25\ MW$$



Neka je $\underline{U}_1 = \underline{U}_1^* = U_1 = 380\ kV$

$$-\underline{E}_1 + \underline{I}_1 j(X_{dG} + X_{T1}) + \underline{U}_1 = 0$$

$$\underline{E}_1 = U_1 + j X_{GT1} \cdot \underline{I}_1 \qquad \underline{I}_1 = \frac{S_1^*}{U_1} = \frac{P_{nat} - \cancel{jQ}}{U_1}$$

$$\underline{E}_1 = U_1 + j(X_{dG} + X_{T1}) \cdot \frac{P_{nat}}{U_1} = 380\ kV + j \cdot 560\ \Omega \cdot \frac{451{,}25\ MW}{380\ kV}$$

$\underline{E}_1 = 380\ kV + j\,265\ kV$

$\underline{E}_1 = 465{,}914\ kV \angle 60{,}255^\circ$

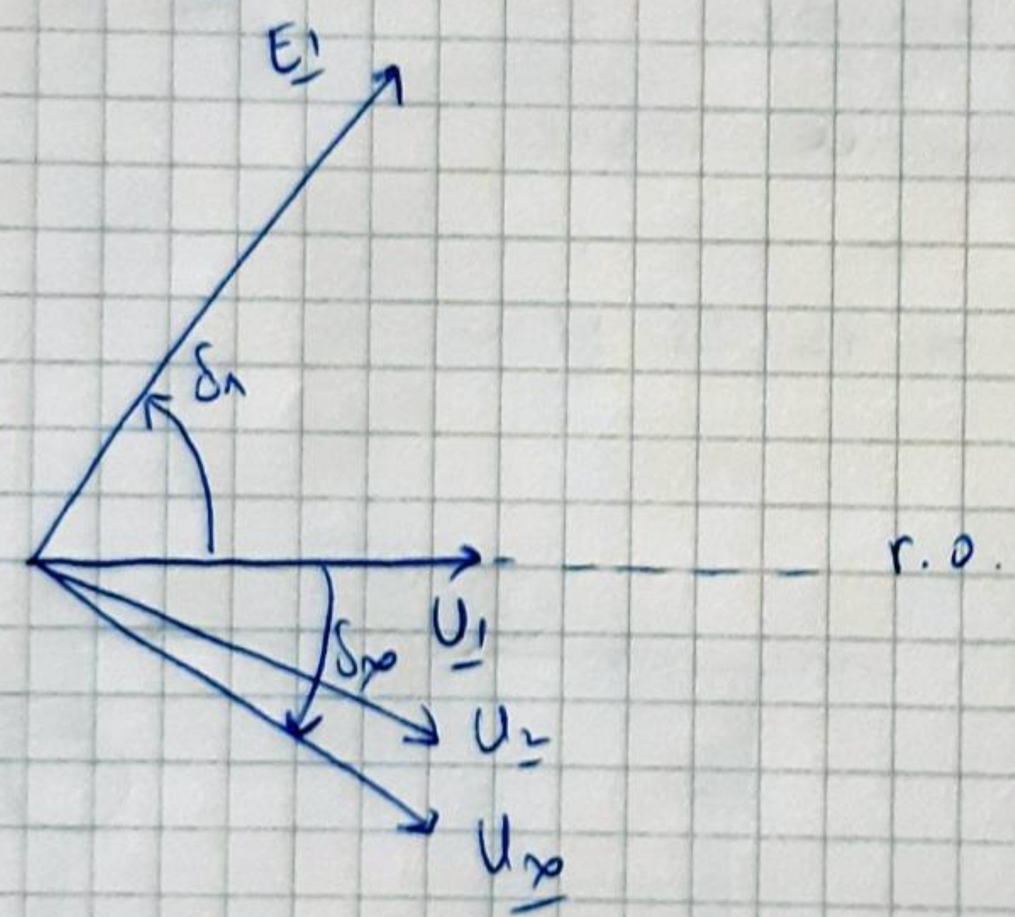
$\underline{U}_1 = \underline{U}_2 \cos\beta l_w + j Z_c \underline{I}_2 \sin\beta l_w$

$\underline{I}_1 = j \dfrac{\underline{U}_2}{Z_c} \sin\beta l_w + \underline{I}_2 \cos\beta l_w$

$P = P_{nat} \Rightarrow Z_c \underline{I}_2 = \underline{U}_2$

$\underline{U}_1 = \underline{U}_2 (\cos\beta l_w + j \sin\beta l_w) = \underline{U}_2 \cdot e^{j\beta l_w}$, $|\underline{U}_1| = |\underline{U}_2|$

$\underline{I}_1 = j \underline{I}_2 \sin\beta l_w + \underline{I}_2 \cos\beta l_w = \underline{I}_2 e^{j\beta l_w}$



$-\underline{U}_2 + \underline{I}_2 \cdot j X_{T2} + \underline{U}_\infty = 0$

$\underline{U}_\infty = \underline{U}_2 - j X_{T2} \underline{I}_2 = \underline{U}_1 e^{-j\beta l_w} - j X_{T2} \underline{I}_1 e^{-j\beta l_w} =$

$= (U_1 - j X_{T2} I_1) e^{-j\beta l_w} = (380 - j\,34{,}295)\, e^{-j\beta l_w}\ kV =$

$= 381{,}544 \angle -(5{,}157^\circ + \beta l_w)$

$$\delta_1 - \delta_2 = 60{,}255^\circ + 5{,}157^\circ + \beta L_V < 90^\circ$$

$$0{,}06\,\frac{^\circ}{km}\, L_V < 24{,}588^\circ$$

$$\boxed{L_V < 409{,}8\ km}$$

Maksimalna udaljenost na koju se može prenijeti prirodna snaga, da bi sistem ostao stabilan na male poremećaje, je $L_{max} = 409{,}8\,km$.

b) Ako se kompenzuje 50% reaktanse voda, ekvivalentna podužna induktivnost je upola manja od podužne induktivnosti voda prije kompenzacije. Ovaj postupak utiče na vrijednost Z_c i β koje se smanjuju $\sqrt{2}$ puta.

$$Z_c = \frac{320\,\Omega}{\sqrt{2}} = 226{,}274\,\Omega$$

$$\beta = \frac{0{,}06\,\frac{^\circ}{km}}{\sqrt{2}} = 0{,}04243\,\frac{^\circ}{km}$$

$P = 451{,}25\ MW$ // prenosi se ista snaga

$$\underline{U}_1 = \underline{U}_2 \cos\beta L_V + j Z_c \underline{I}_2 \sin\beta L_V$$

$$\underline{I}_1 = j\frac{\underline{U}_2}{Z_c}\sin\beta L_V + \underline{I}_2 \cos\beta L_V$$

$P = 451{,}25\ MW$ nije prirodna snaga pa $\underline{U}_2 \neq Z_c \underline{I}_2$

$$\begin{bmatrix} \underline{U}_1 \\ \underline{I}_1 \end{bmatrix} = \begin{bmatrix} \cos\beta L_w & j Z_c \sin\beta L_w \\ j\frac{1}{Z_c}\sin\beta L_w & \cos\beta L_w \end{bmatrix} \begin{bmatrix} \underline{U}_2 \\ \underline{I}_2 \end{bmatrix}$$

$$\underline{I}_2 = \underline{I}_\infty$$

$$\underline{U}_\infty = \underline{U}_2 - jX_{T2}\underline{I}_2 \quad , \quad \underline{U}_2 = \underline{U}_\infty + jX_{T2}\underline{I}_\infty$$

$$\begin{bmatrix} \underline{U}_2 \\ \underline{I}_2 \end{bmatrix} = \begin{bmatrix} 1 & jX_{T2} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \underline{U}_\infty \\ \underline{I}_\infty \end{bmatrix}$$

$$\begin{bmatrix} \underline{U}_1 \\ \underline{I}_1 \end{bmatrix} = \begin{bmatrix} \cos\beta L_w & jZ_c\sin\beta L_w \\ j\frac{1}{Z_c}\sin\beta L_w & \cos\beta L_w \end{bmatrix} \begin{bmatrix} 1 & jX_{T2} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \underline{U}_\infty \\ \underline{I}_\infty \end{bmatrix} =$$

$$= \begin{bmatrix} \cos\beta L_w & j(X_{T2}\cos\beta L_w + Z_c\sin\beta L_w) \\ j\frac{1}{Z_c}\sin\beta L_w & \cos\beta L_w - \frac{1}{Z_c}X_{T2}\sin\beta L_w \end{bmatrix} \begin{bmatrix} \underline{U}_\infty \\ \underline{I}_\infty \end{bmatrix}$$

$$\begin{bmatrix} \underline{U}_\infty \\ \underline{I}_\infty \end{bmatrix} = \begin{bmatrix} \cos\beta L_w - \frac{1}{Z_c}X_{T2}\sin\beta L_w & -j(X_{T2}\cos\beta L_w + Z_c\sin\beta L_w) \\ -j\frac{1}{Z_c}\sin\beta L_w & \cos\beta L_w \end{bmatrix} \begin{bmatrix} \underline{U}_1 \\ \underline{I}_1 \end{bmatrix}$$

$$\underline{U}_\infty = U_1\left[\cos\beta L_w - \frac{1}{Z_c}X_{T2}\sin\beta L_w\right] - jI_1\left[X_{T2}\cos\beta L_w + Z_c\sin\beta L_w\right]$$

$U_1 = 380$ kV

$$I_1 = \frac{P}{U_1} = 1187{,}5\,A$$

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$$\underline{U}_\infty = 380(\cos\beta L_v - 0{,}1276\sin\beta L_v) - j(34{,}295\cos\beta L_v + 268{,}7\sin\beta L_v)$$

$$\delta_\infty = -\operatorname{arctg}\frac{34{,}295\cos\beta L_v + 268{,}7\sin\beta L_v}{380\cos\beta L_v - 48{,}488\sin\beta L_v}$$

$$\underline{E}_1 = \underline{U}_1 + jX_{GT1}\underline{I}_1 = U_1 + jX_{GT1}I_1 = 765{,}914\angle 60{,}255^\circ \text{ kV}$$

$$\delta_1 = 60{,}255^\circ$$

$$\delta_1 + \operatorname{arctg}\frac{34{,}295\cos\beta L_v + 268{,}7\sin\beta L_v}{380\cos\beta L_v - 48{,}488\sin\beta L_v} < 90^\circ$$

$$\operatorname{arctg}\frac{34{,}295\cos\beta L_v + 268{,}7\sin\beta L_v}{380\cos\beta L_v - 48{,}488\sin\beta L_v} < 29{,}745^\circ$$

$$34{,}295\cos\beta L_v + 268{,}7\sin\beta L_v < 217{,}144\cos\beta L_v - 27{,}407\sin\beta L_v$$

$$296{,}407\sin\beta L_v < 182{,}849\cos\beta L_v$$

$$\operatorname{tg}\beta L_v < 0{,}61688$$

$$\beta L_v < 31{,}67^\circ$$

$$\boxed{L_v < 746{,}4\text{ km}}$$

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